

6ai	When light intensity is maximum, resistance of LDR = 1200 Ω	C1

	Total resistance = $\frac{1}{\left(\frac{1}{1200}\right) + \left(\frac{1}{600}\right)} = 400 \, \Omega$	A1
6aii	For minimum p.d. across R_2 , $R_1 = 400 \, \Omega$ total parallel resistance ($R_2 + \text{LDR}$) is lower than R_2 (minimum) p.d. across R_2 in Fig. 6.1 is lower than that in Fig. 6.2	M1 M1 A1
6bi	At balance length, no current in E_1 or r , so $E_1 = V_{XY}$ (Balance length $XY = 100.0 - 37.5 = 62.5 \, \text{cm}$) (Using potential divider principle,) $V_{XZ} = \left(\frac{R_{XZ}}{R_{XZ} + R_1} \right) E = \left(\frac{10.0}{10.0 + 15.0} \right) (2.0) = 0.80 \, \text{V}$ $V_{XY} = \frac{L_{XY}}{L_{XZ}} (V_{XZ}) = \left(\frac{62.5}{100.0} \right) (0.80) = 0.50 \, \text{V}$ Therefore $E_1 = V_{XY} = 0.50 \, \text{V}$	M1 M1 M1 A0
6bii	$V_{XY} = V_{R_2}$ $\frac{10.0}{100.0} \times 2.0 = \frac{R_2}{R_2 + r} E_1$ $= \frac{5.0}{5.0 + r} \times 0.50$ $0.20(5.0 + r) = 2.5$ $r = \frac{2.5}{0.20} - 5.0 = 12.5 - 5.0 = 7.5 \, \Omega$	C1 C1 A1

7ai	0.6	A1
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